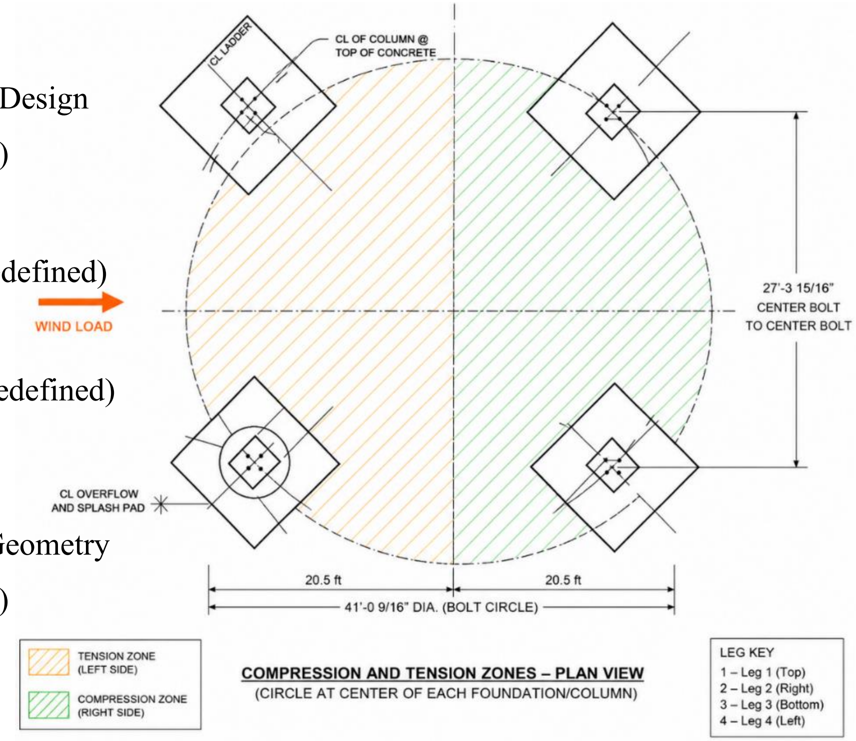
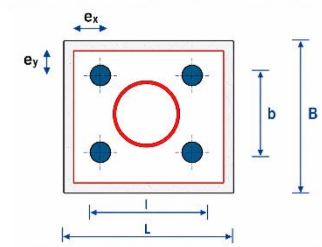
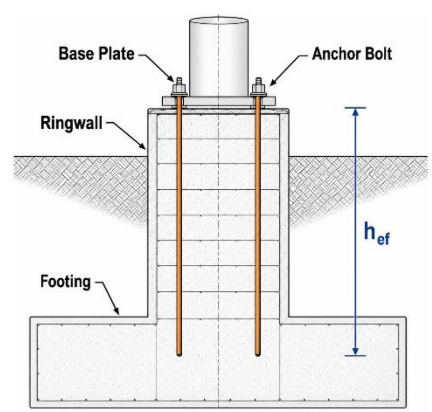
1. Anchor Bolts segment

Design Parameter	Symbol	Value	Source
Factored axial load (LRFD)	P_u	1609.756 kips	Main Tank Design (Predefined)
Factored moment (LRFD)	M_u	5673.437 kip-ft	Main Tank Design (Predefined)
Factored shear force (LRFD)	V_u	52.30 kips	Main Tank Design (Predefined)
Number of anchor bolts	N_b	4	User Input
Diameter circle representing axisymmetric spacing between legs/columns	D_{cl}	41′-0.5625″	MainTankGeometry (Predefined)
Anchor bolt diameter	d_b	1.75 in	User Input
Effective embedment depth	h_{ef}	40 in	User Input
Concrete compressive strength	f'_c	4000 psi	User Input
Anchor bolt yield strength	F_y	36 ksi	User Input
Anchor bolt ultimate tensile strength	F_u	58 ksi	User Input
Edge distance	$e_x = e_y$	13.5 in	AISC Prescribed Minimum
Pedestal dimensions	B x L	39 in × 39 in	User Input
Bolt arrangement dimensions	b x l	12 in × 12 in	From the Pedestal Design
Strength reduction factors (tension, shear)	ϕ_t, ϕ_v	0.75, 0.75	ACI 318-19







STEP 1 — DETERMINE BOLT ARRANGEMENT DIMENSIONS (b x l)

e_x .... Extracting from the table below.
e_y .... Extracting from the table below.

Pedestal:

$39''$

Bolt spacing:

$12''$

Edge distance:

$$\frac{39-12}{2} = 13.5''$$

Required minimum:

$$7d = 7(1.75) = 12.25''$$

$b = B - 2 * e_x = 39 - 2(13.5) = 12$ in
$l = L - 2 * e_y = 39 - 2(13.5) = 12$ in

STEP 2 — DETERMINE TENSION SIDE LEGS

For overturning, assume:

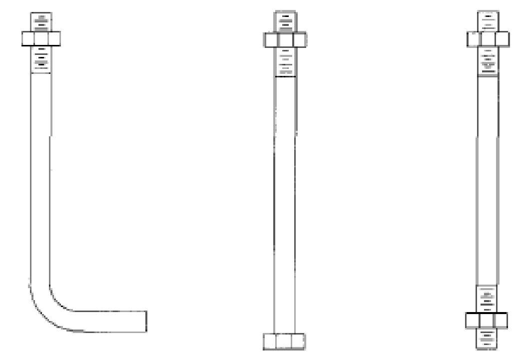
- 2 legs in compression
- 2 legs in tension

Thus…………………………………………………**nt=2**

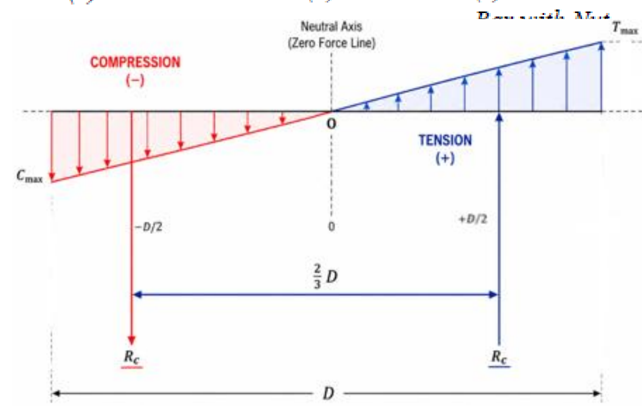
Minimum Bolt Lengths and Edge Distances

Shipp and Haninger (1983) have presented minimum guidelines for bolt embedment and edge distance, adopted from ACI 349. These are presented for use in designing anchor bolts for tension as follows:

Bolt Type, Material	Minimum Embedded Length	Minimum Embedded Edge Distance
A307, A36	12 *d*	5 *d* > 4 in.
A325, A449	17 *d*	7 *d* > 4 in.

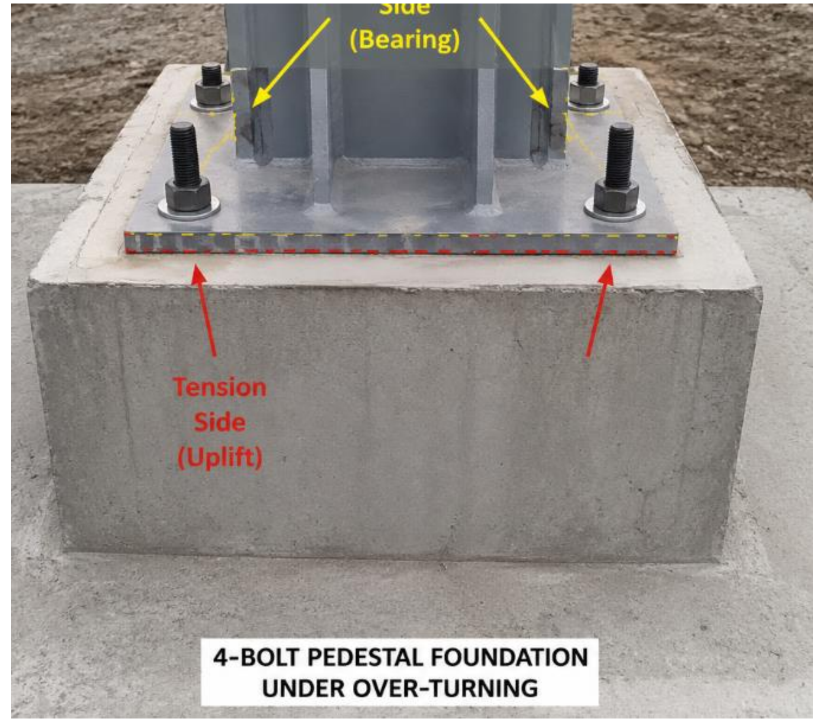


(a) Hooked Bar (b) Bolt (c) Threaded



CRITICAL SECTION (DIAMETRAL SECTION)

STEP 3 — DETERMINE LEG RADIUS



$$D = 41' - 0\frac{9}{16}"$$

Radius:

$$r = 20.523'$$

Convert:

$$r = 246.28"$$

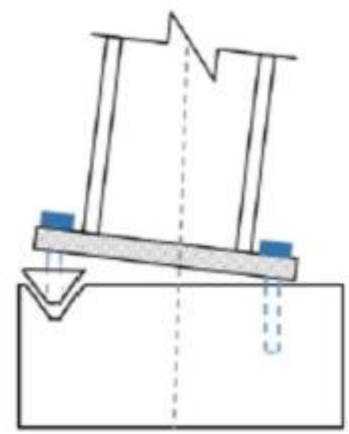
STEP 4 — TENSION FORCE FROM OVERTURNING

$$T = \frac{M_u}{4/3r}$$

Convert moment:

$$M_u = 5673.44 \times 12$$

$$M_u = 68081.3 \text{ kip-in}$$



Anchor bolt failure

Now:

$$T = \frac{68081.3}{\frac{4}{3} * 246.28}$$

$$T = 207.3\ kips$$

This is total overturning uplift.

Shared by 2 tension legs:

$$T_{leg} = \frac{207.3}{2}$$

$$T_{leg} = 103.65\ kips$$

STEP 5 — TENSION PER BOLT

$$T_{bolt} = \frac{103.65}{4}$$

$$T_{bolt} = 25.91\ kips$$

STEP 6 — REQUIRED STEEL AREA

LRFD:

$$\phi = 0.75$$

Required steel area:

$$A_s = \frac{T_u}{\phi F_y}$$

$$A_s = \frac{25.91}{0.75(36)}$$

$$A_s = 0.96\ \text{in}^2$$

STEP 7 — SELECT BOLT DIAMETER

Diameter	Area
1.75 in	1.90 in^2
2.0 in	2.50 in^2
2.25 in	3.25 in^2

Now:

$$A_s = 0.96\ in^2$$

SELECT

1.75" bolt is sufficient

Area:

$$A_b = 1.90\ \text{in}^2$$

STEP 8 — CHECK STEEL TENSION CAPACITY

$$\phi N_s = \phi A_b F_y$$
$$\phi N_s = 0.75(1.90)(36)$$
$$\phi N_s = 51.3 \text{ kips}$$

Demand

25.91 kips

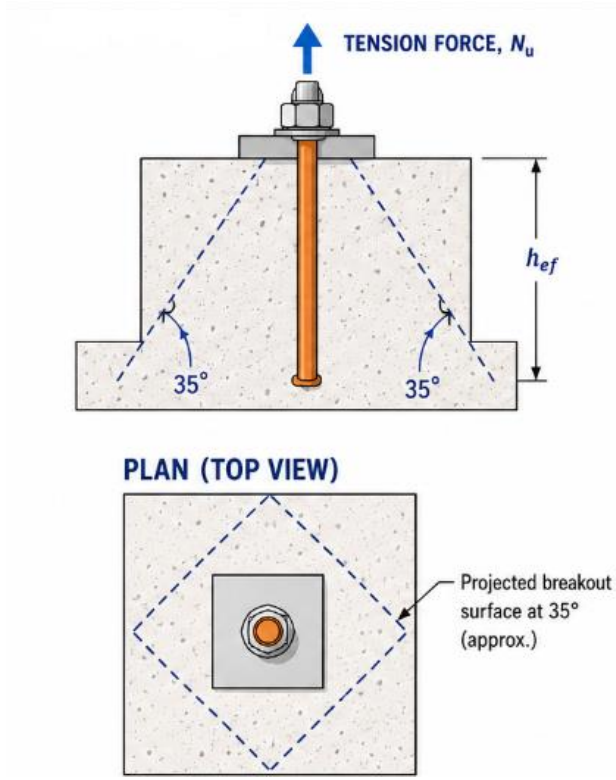
Check:

$51.3 > 25.91$………………**PASS**

STEP 9 — CHECK STEEL SHEAR CAPACITY

Shear per pedestal:

$$V_{bolt} = \frac{V_u}{N_b \; x \; N_{leg}}$$
$$V_{bolt} = \frac{52.3}{4x4}$$
$$V_{bolt} = 3.27 \text{ kips}$$



Steel shear capacity:

$$\phi V_s = 0.75(0.6)(36)(1.90)$$
$$\phi V_s = 30.78 \text{ kips}$$

Check:

$30.78 > 3.27$……………………..**PASS**

STEP 10 — CONCRETE BREAKOUT CHECK

ACI approximate breakout: (17.6.2 and 17.3.3).

Add a note adjacent to the input box indicating the required information. Also, display the corresponding Table X near the input box for user reference.

$$N_{cb} = \phi k_c \sqrt{f_c'} h_{ef}^{1.5}$$

Use:

- $k_c = 24$
- $\phi = 0.75$

Bolt Type, Material	Minimum Embedded Length	Minimum Embedded Edge Distance
A307, A36	12 *d*	5 *d* > 4 in.
A325, A449	17 *d*	7 *d* > 4 in.

Compute:

$$\sqrt{4000} = 63.25$$

$$40^{1.5} = 253$$

$$N_{cb} = 0.75(24)(63.25)(253)$$

$$N_{cb} = 288000 \text{ lb}$$

$$N_{cb} = 288 \text{ kips}$$

Table 17.5.3(a)—Anchor strength governed by steel

Type of steel element	Strength reduction factor ϕ	
	Tension (steel)	Shear (steel)
Ductile	0.75	0.65
Brittle	0.65	0.60

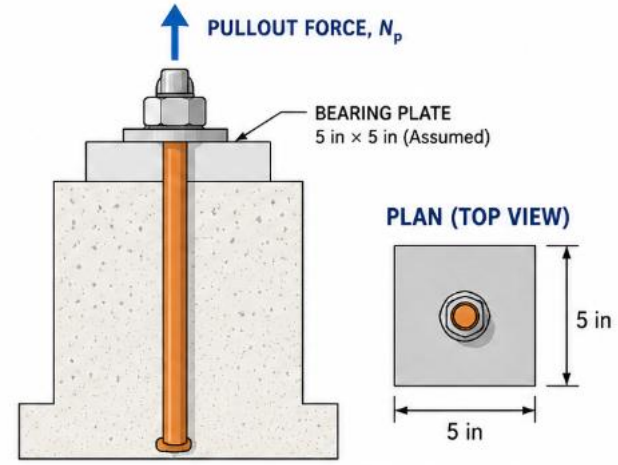
Check:

$288 > 25.91$……………………**PASS**

STEP 11 — PULLOUT CHECK

Pullout strength shall be verified using the actual anchor head or bearing plate dimensions provided in the final anchor detail. For preliminary design, a 5 in × 5 in bearing plate was assumed.

$$A_{brg} = 25 \text{ in}^2$$



Pullout:

$$N_p = 8A_{brg}f_c'$$

A_{brg}= bearing area of anchor head or plate washer (Assume a 5 in × 5 in square plate washer)……..

ACI 318 Chapter 17 requires pullout strength to be based on the actual anchor geometry

$$N_p = 8(25)(4000)$$

$$N_p = 800000 \text{ lb}$$

$$N_p = 800 \text{ kips}$$

Check:

$$800 > 34.55 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$

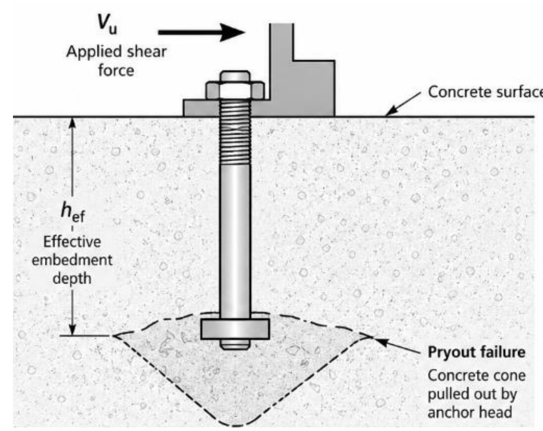
STEP 12 — PRYOUT CHECK

ACI 318-19 Chapter 17 (Anchoring to Concrete)

$$V_{cp} = k_{cp} * N_{cb}$$

where:

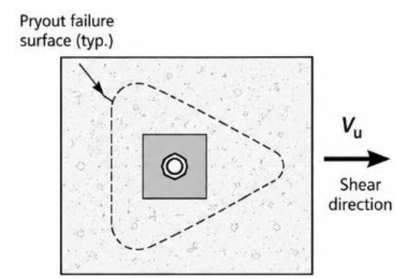
- V_{cp}= pryout strength
- N_{cb}= concrete breakout strength in tension
- k_{cp}= pryout factor



k_{cp} **= 1.0 for** h_{ef} **< 2.5 in.**

k_{cp} **= 2.0 for** h_{ef} **≥ 2.5 in.**

ACI 318-19 Table 17.7.3.1 (Pryout Strength Factor k_{cp})



$$V_{cp} = 2.0N_{cb}$$

$$V_{cp} = 576 \text{ kips}$$

Check:

$$576 > 3.27 kips \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$

STEP 13 — INTERACTION CHECK

$$\left(\frac{N_u}{\phi N_n}\right)^2 + \left(\frac{V_u}{\phi V_n}\right)^2 \leq 1.0$$

Compute:

$$\left(\frac{34.55}{51.30}\right)^2 + \left(\frac{3.27}{30.78}\right)^2$$

$$= 0.454 + 0.011$$

$$= 0.465$$

Check: $0.465 < 1.0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$ **PASS**